ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At most 12" excludes 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

3201

QUESTION ID

S86-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Includes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 1

Satisfy several constraints at once • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Which whole-number values of x give a volume of at least 500 cm³ AND a height of at most 3 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

3203

QUESTION ID

S86-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2 and x = 3

GRADING NOTE Form required: both values. Volumes 512 and 588; x = 4, 5 are too tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ramp must rise 2 ft and fit within 20 ft of horizontal space. Can it meet the 1 : 12 limit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

3204

QUESTION ID

S86-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: it needs 24 ft of run.

GRADING NOTE Form required: the conclusion with the run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

True or false?

A design that satisfies one constraint may fail another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

3205

QUESTION ID
S86-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Check all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

A rectangular garden needs area 48 m² and width at most 6 m. Which works?

A. 6 m wide by 8 m long

B. 4 m wide by 10 m long

C. 7 m wide by 7 m long

D. 8 m wide by 6 m long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

3206

QUESTION ID

S86-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6 m wide by 8 m long

GRADING NOTE Width 6, area 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder 3 m tall must hold at least 10 m³ and stand on a 2 m by 2 m floor space. Is that possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

3207

QUESTION ID

S86-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the largest is $r = 1$ m, giving $3\pi \approx 9.4$ m³ < 10 m³.

GRADING NOTE Form required: the conclusion with the volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 2

Find the best design with a function graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Write its volume as a function V(x) and state the domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

3209

QUESTION ID

S86-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$V(x) = x(20 - 2x)^2$, $0 < x < 10$

GRADING NOTE Form required: the function and the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An open box made from a 20 cm by 20 cm sheet with x-cm corner squares cut out has volume $V(x) = x(20 - 2x)^2$. Use a graph or table to estimate the x that gives the largest volume, to the nearest tenth, and that volume to the nearest cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

3210

QUESTION ID
S86-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x \approx 3.3$; $V \approx 593\ cm^3$

GRADING NOTE Form required: both. (Exact: x = 10/3, V = 16,000/27 ≈ 592.6.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

True or false?

$V(x) = x(20 - 2x)^2$ has zeros at $x = 0$ and $x = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

3211

QUESTION ID

S86-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The zeros are $x = 0$ and $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet, $V(x) = x(20 - 2x)^2$. The graph of V on $0 < x < 10$ rises and then falls. Its highest point gives

A. the cut that makes no box

B. the base area

C. the largest volume

D. the smallest volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

3212

QUESTION ID
S86-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. the largest volume

GRADING NOTE Maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet with x-cm corner cuts, $V(x) = x(20 - 2x)^2$ and V is largest near x = 3.3. Why is x = 5 not the best cut even though it is halfway between 0 and 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

3213

QUESTION ID
S86-F2-Q5

TYPE
Short answer

CORRECT ANSWER
V(5) = 500 < 593; the maximum of a cubic need not be at the middle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

SET 87 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to minimize cost and work with typographic grids

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Surface area of prisms and cylinders	**7.G.B.6**
BACKWARD 2	Proportional reasoning with ratios	**7.RP.A.2**
BACKWARD 3	Unit prices and rounding up purchases	**7.EE.B.3**
FORWARD 1	Compare designs with equal volume for least material	**HSG-MG.A.3**
FORWARD 2	Decide between designs using expected cost	**HSS-MD.B.7**

SET 87 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A can with radius 4 cm and height 10 cm is made of metal costing \$0.002 per cm². Find the area of metal (top, bottom, and side) and the cost, to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

3216

QUESTION ID

S87-M-Q1

TYPE

Short answer

CORRECT ANSWER

112π cm² ≈ 351.9 cm²; cost \$0.70

GRADING NOTE Form required: the area and the cost. $2\pi r^2 + 2\pi rh = 32\pi + 80\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Can A has r = 4 cm, h = 10 cm. Can B has r = 5 cm, h = 6.4 cm. Show that they hold the same volume, then decide which uses less metal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

3217

QUESTION ID
S87-M-Q2

TYPE
Short answer

CORRECT ANSWER
Both hold 160π cm³; A uses 112π cm², B uses 114π cm², so A uses less metal.

GRADING NOTE Form required: both volumes, both areas, and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A wall 12 m by 3 m has a window 1.5 m by 1.2 m. Paint covers 10 m² per liter and is sold only in whole liters at $8 each. Find the cost to paint the wall once.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

3218

QUESTION ID
S87-M-Q3

TYPE
Short answer

CORRECT ANSWER
$32

GRADING NOTE Any equivalent form. 34.2 m² needs 3.42 L, so buy 4 L.

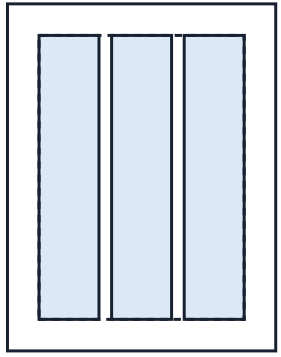
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A page is 8.5 in. wide with 1 in. margins on each side. The text area has 3 equal columns separated by 2 gutters of 0.2 in. Find the width of each column to the nearest hundredth of an inch.

Width: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

3219

QUESTION ID

S87-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.03 in.

GRADING NOTE Rounding to 0.01 required. (6.5 − 0.4) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A poster grid uses the golden ratio (about 1.618 : 1) for height : width. A poster is 60 cm wide. Its height is about

A. 120 cm

B. 97.1 cm

C. 37.1 cm

D. 61.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

3220

QUESTION ID
S87-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. 97.1 cm

GRADING NOTE 60 × 1.618.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 1

Surface area of prisms and cylinders • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of the two circular ends of a can with radius 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

3222

QUESTION ID

S87-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of the side of a can with radius 4 and height 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

3223

QUESTION ID

S87-B1-Q2

TYPE

Short answer

CORRECT ANSWER

80π

GRADING NOTE Form required: exact value. 2πrh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

Surface area of a 2 × 3 × 4 box?

A. 24

B. 48

C. 52

D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

3224

QUESTION ID

S87-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 52

GRADING NOTE 2(6 + 8 + 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

True or false?

The side of a cylinder unrolls into a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

3225

QUESTION ID

S87-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Net.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

True or false?

A cylinder's surface area is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

3226

QUESTION ID

S87-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 2

Proportional reasoning with ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Height : width = 1.618 : 1 and width = 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q1/5

3228

QUESTION ID

S87-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16.18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Split 6.1 in. into 3 equal columns. Width of each (nearest hundredth)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q2/5

3229

QUESTION ID

S87-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.03 in.

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 2 : 3 rectangle 30 cm wide (width = 2 parts) is

A. 45 cm tall

B. 60 cm tall

C. 20 cm tall

D. 33 cm tall

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q3/5

3230

QUESTION ID

S87-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45 cm tall

GRADING NOTE 15 per part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q3/5

True or false?

Equal columns divide the available width equally.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q4/5

3231

QUESTION ID

S87-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 : 2 ratio means the second is half the first.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q5/5

3232

QUESTION ID

S87-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 3

Unit prices and rounding up purchases • Grade 7 • Expressions & Equations • 7.EE.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Paint needed is 3.42 L, sold in whole liters. How many liters must be bought?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

3234

QUESTION ID

S87-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

3234

ANSWER KEY Reduced copy of the question page, with its answer

Metal costs $0.002 per cm². Cost of 350 cm²?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

3235

QUESTION ID

S87-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$0.70

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Fabric is $6 per yard, sold in whole yards. You need 4.2 yd. Cost?

A. $30

B. $25.20

C. $24

D. $26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

3236

GRADING NOTE Buy 5 yd.

QUESTION ID

S87-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

3236

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When items are sold in whole units, round the amount needed up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

3237

QUESTION ID

S87-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Enough material.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3.1 cans of paint can be bought as 3 cans.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

3238

QUESTION ID

S87-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Need 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 1

Compare designs with equal volume for least material • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A box with a square base must hold 1,000 cm³. Compare the surface areas for base sides 8, 10, and 12 cm. Which uses the least material?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

3240

QUESTION ID

S87-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 cm: 628 cm²; 10 cm: 600 cm²; 12 cm: about 621 cm²; the 10 cm cube uses the least

GRADING NOTE Form required: the three areas and the choice. h = 1000/s².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 12-column grid on a 6.5 in. text area has 11 gutters of 0.125 in. Find the width of each column to the nearest thousandth of an inch.

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

3241

QUESTION ID

S87-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.427 in.

GRADING NOTE Rounding to 0.001 required. (6.5 − 1.375) ÷ 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

True or false?

Among boxes with a square base and a fixed volume, the cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

3242

QUESTION ID

S87-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Classic result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 12-column grid can also be used as

A. only 12 columns

B. 7 equal columns

C. 5 equal columns

D. 4 columns of 3, 3 of 4, or 2 of 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

3243

QUESTION ID
S87-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 4 columns of 3, 3 of 4, or 2 of 6

GRADING NOTE Divisors of 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do designers often choose 12-column grids?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

3244

QUESTION ID

S87-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12 divides evenly into 2, 3, 4, and 6, giving many layout options.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 2

Decide between designs using expected cost • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Packaging A costs $1.20 per item and 2% of items break. Packaging B costs $0.90 and 8% break. A break costs $20. Find the expected cost per item for each and choose.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

3246

QUESTION ID
S87-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A: $1.20 + 0.02 × $20 = $1.60; B: $0.90 + 0.08 × $20 = $2.50; choose A

GRADING NOTE Form required: both expected costs and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A design uses a cheaper metal that fails 5% of the time, with replacement cost $40. How much cheaper must it be per can to be worth it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

3247

QUESTION ID
S87-F2-Q2

TYPE
Short answer

CORRECT ANSWER
More than $2.00 cheaper (0.05 × $40)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

True or false?

The cheapest material always gives the cheapest design.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

3248

QUESTION ID
S87-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Failures and replacement costs can make it more expensive overall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

Expected cost =

A. upfront cost × probability

B. upfront cost + probability of failure × cost of failure

C. cost of failure only

D. upfront cost − probability

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

3249

QUESTION ID
S87-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. upfront cost + probability of failure × cost of failure

GRADING NOTE Weighted cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name one cost besides material that a can designer might consider.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

3250

QUESTION ID
S87-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example shipping weight, labor, breakage, or recycling.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9 • ADVANCED EXTENSION (+)

Derive and use the formula A = ½ab sin C

MAIN	Geometry main question	**HSG-SRT.D.9**
BACKWARD 1	The sine ratio in a right triangle	**HSG-SRT.C.6**
BACKWARD 2	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 3	The included angle of two sides	**HSG-CO.B.8**
FORWARD 1	Solve for a side or an angle from an area	**HSG-SRT.D.9**
FORWARD 2	From equal areas to the Law of Sines	**HSG-SRT.D.10**

SET 88 • HSG-SRT.D.9

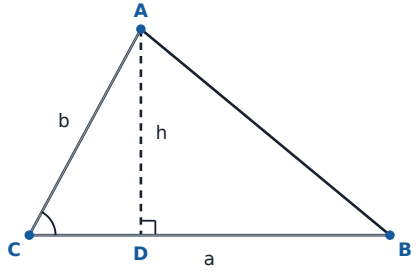
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, the altitude from A to side CB has length h. Derive the formula Area = ½ab sin C.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

3253

QUESTION ID

S88-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

h = b sin C, so Area = ½ah = ½ab sin C.

GRADING NOTE REQUIRED WORK: Express h using sin C in right triangle ADC, then use Area = ½ · base · height.

Any valid proof (paragraph, two-column, or flow). Essential: in right triangle ADC, sin C = h/b, so h = b sin C; Area = ½ · a · h; substitute. Alternative: an altitude from B gives the same result with h = a sin C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

A triangle has sides a = 8 and b = 11 with included angle C = 40°. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

3254

QUESTION ID

S88-M-Q2

TYPE

Short answer

CORRECT ANSWER

28.3

GRADING NOTE Rounding to 0.1 required. ½ · 8 · 11 · sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two sides of a triangle are 6 and 10, and the angle between them is 150°. Find the exact area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

3255

QUESTION ID
S88-M-Q3

TYPE
Short answer

CORRECT ANSWER
15

GRADING NOTE Any equivalent form. sin 150° = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has sides 7 cm and 12 cm with an angle of 55° between them. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

3256

QUESTION ID

S88-M-Q4

TYPE

Short answer

CORRECT ANSWER

68.8 cm²

GRADING NOTE Rounding to 0.1 required. Two congruent triangles: 7 · 12 · sin 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

A triangle has sides 5 and 9 with an included angle of 30°. Its area is

A. 45

B. 22.5

C. 11.25

D. 19.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

3257

QUESTION ID

S88-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 11.25

GRADING NOTE ½ · 45 · ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 1

The sine ratio in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, sin of an acute angle = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

3259

QUESTION ID

S88-B1-Q1

TYPE

Short answer

CORRECT ANSWER

opposite ÷ hypotenuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 10, angle 30°. Find the side opposite the angle.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

3260

QUESTION ID

S88-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin C = h/b, then h =

A. b + sin C

B. sin C/b

C. b/sin C

D. b sin C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

3261

QUESTION ID

S88-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b sin C

GRADING NOTE Multiply by b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 90° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

3262

QUESTION ID

S88-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin C = adjacent ÷ hypotenuse

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

3263

QUESTION ID
S88-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That is cos C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 2

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Base 8, height 5: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

3265

QUESTION ID

S88-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

True or false?

Area = ½ · base · height

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

3266

QUESTION ID

S88-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

The height must be ___ to the base.

A. equal

B. adjacent

C. parallel

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

3267

QUESTION ID

S88-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. perpendicular

GRADING NOTE Altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Base a and height b sin C: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

3268

QUESTION ID

S88-B2-Q4

TYPE

Short answer

CORRECT ANSWER

½ab sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

True or false?

A slanted side can be used as the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

3269

QUESTION ID

S88-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 3

The included angle of two sides • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In triangle ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

3271

QUESTION ID

S88-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠A

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

True or false?

SAS uses the angle between the two given sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

3272

QUESTION ID

S88-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For sides a (BC) and b (CA), the included angle is

A. C

B. none

C. A

D. B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

3273

QUESTION ID

S88-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. C

GRADING NOTE Shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why must the angle be included for ½ab sin C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

3274

QUESTION ID

S88-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The altitude is built from side b and the angle between b and the base a.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any angle of the triangle works in ½ab sin C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

3275

QUESTION ID

S88-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the included one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 1

Solve for a side or an angle from an area • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 10 and 12 and area 30. Find all possible measures of the included angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

3277

QUESTION ID

S88-F1-Q1

TYPE

Short answer

CORRECT ANSWER

30° or 150°

GRADING NOTE Form required: both. sin C = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has area 50, one side 20, and an included angle of 45°. Find the other side to the nearest tenth.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

3278

QUESTION ID

S88-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.1

GRADING NOTE Rounding to 0.1 required. 50 = ½ · 20 · b · sin 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

True or false?

Two different triangles can have the same two sides and the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

3279

QUESTION ID
S88-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE C and 180° − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The largest area of a triangle with sides 6 and 8 occurs when the included angle is

A. 60°

B. 180°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

3280

QUESTION ID

S88-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 90°

GRADING NOTE sin C = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of an equilateral triangle with side 6 using ½ab sin C (exact).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

3281

QUESTION ID
S88-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$9\sqrt{3}$

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 2

From equal areas to the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

The area of triangle ABC can be written as ½bc sin A, ½ac sin B, or ½ab sin C. Use this to show that sin A/a = sin B/b = sin C/c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

3283

QUESTION ID

S88-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

½bc sin A = ½ac sin B = ½ab sin C; dividing by ½abc gives sin A/a = sin B/b = sin C/c.

GRADING NOTE REQUIRED WORK: Set the three expressions equal and divide by ½abc.

Any valid proof (paragraph, two-column, or flow). Essential: three area expressions equal; division by ½abc; result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use sin A/a = sin B/b: A = 30°, a = 5, b = 8. Find sin B.

sin B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

3284

QUESTION ID

S88-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

True or false?

The Law of Sines can be derived from the area formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

3285

QUESTION ID
S88-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE As above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

Dividing ½bc sin A = ½ac sin B by ½abc gives

A. a sin A = b sin B

B. sin A/a = sin B/b

C. sin A = sin B

D. a/b = sin A sin B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

3286

QUESTION ID

S88-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. sin A/a = sin B/b

GRADING NOTE Cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is it allowed to divide by ½abc?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

3287

QUESTION ID

S88-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Side lengths are positive, so ½abc ≠ 0.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Sines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	Use trigonometry in right triangles	**HSG-SRT.C.8**
BACKWARD 2	The area formula ½ab sin C	**HSG-SRT.D.9**
BACKWARD 3	Proportions and cross-multiplying	**7.RP.A.2**
FORWARD 1	Use the Law of Sines (AAS and ASA)	**HSG-SRT.D.10**
FORWARD 2	Surveying with the Law of Sines	**HSG-SRT.D.11**

SET 89 • HSG-SRT.D.10

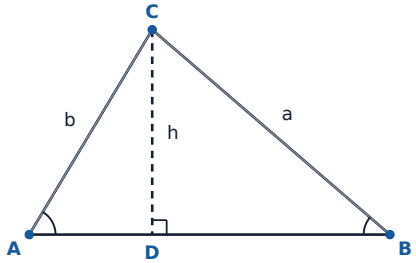
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, CD is the altitude to AB. Prove that a/sin A = b/sin B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

3290

QUESTION ID

S89-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

h = b sin A and h = a sin B, so b sin A = a sin B and a/sin A = b/sin B.

GRADING NOTE REQUIRED WORK: Write h two ways, using right triangles ADC and BDC.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = h/b and sin B = h/a from the two right triangles; equate h; divide by sin A sin B. Repeating with another altitude gives c/sin C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has an obtuse angle at A, so the altitude from C meets line AB extended past A at D. Prove that a/sin A = b/sin B in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

3291

QUESTION ID

S89-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

h = b sin(180° − A) = b sin A and h = a sin B, so a/sin A = b/sin B.

GRADING NOTE REQUIRED WORK: Use the right triangle with angle 180° − A at A, and sin(180° − A) = sin A.

Any valid proof (paragraph, two-column, or flow). Essential: the exterior angle at A is 180° − A; h = b sin(180° − A); the identity sin(180° − A) = sin A; equate with h = a sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the area formula ½bc sin A = ½ac sin B gives the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

3292

QUESTION ID

S89-M-Q3

TYPE

Short answer

CORRECT ANSWER

Dividing both sides by ½abc gives sin A/a = sin B/b.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the proof of the Law of Sines, the altitude from C to AB has length h and forms right triangle ADC. Which fact gives h = b sin A?

A. The triangle angle sum

B. sin A = opposite/hypotenuse in right triangle ADC

C. cos A = h/b

D. The Pythagorean Theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

3293

QUESTION ID
S89-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. sin A = opposite/hypotenuse in right triangle ADC

GRADING NOTE Right triangle ADC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

True or false?

The Law of Sines works only in right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

3294

QUESTION ID

S89-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It works in every triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 1

Use trigonometry in right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle: hypotenuse 12, angle 40°. Opposite side to the nearest tenth?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

3296

QUESTION ID

S89-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

True or false?

An altitude splits a triangle into two right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

3297

QUESTION ID

S89-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin B = h/a, then h =

A. h sin B

B. a/sin B

C. a sin B

D. sin B/a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

3298

QUESTION ID

S89-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a sin B

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle: opposite 5, hypotenuse 10. Angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

3299

QUESTION ID
S89-B1-Q4

TYPE
Short answer

CORRECT ANSWER
30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

True or false?

sin is defined only for angles in right triangles, so it cannot be used after drawing an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

3300

QUESTION ID

S89-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The altitude creates right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 2

The area formula ½ab sin C • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has b = 4, c = 5, and A = 90°. What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

3302

QUESTION ID

S89-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

True or false?

½bc sin A, ½ac sin B, and ½ab sin C all give the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

3303

QUESTION ID

S89-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Area of a triangle with sides 6, 6 and included 30°?

A. 3

B. 9

C. 36

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

3304

GRADING NOTE ½ · 36 · ½.

QUESTION ID

S89-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the area of triangle ABC using sides a and c.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

3305

QUESTION ID

S89-B2-Q4

TYPE

Short answer

CORRECT ANSWER

½ac sin B

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

½ab sin C uses angle A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

3306

QUESTION ID

S89-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angle C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 3

Proportions and cross-multiplying • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/6 = 10/4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

3308

QUESTION ID

S89-B3-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

True or false?

If b sin A = a sin B, then a/sin A = b/sin B (when the sines are not 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

3309

QUESTION ID

S89-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by sin A sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From 3x = 4y, x/y =

A. 3/4

B. 4/3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

3310

QUESTION ID

S89-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/3

GRADING NOTE Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite p · q = r · s as a proportion with p and r on top.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

3311

QUESTION ID

S89-B3-Q4

TYPE

Short answer

CORRECT ANSWER

p/r = s/q (or p/s = r/q)

GRADING NOTE Any correct proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

a/b = c/d means a + d = b + c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

3312

QUESTION ID

S89-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 1

Use the Law of Sines (AAS and ASA) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, B = 60°, and a = 10. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

3314

QUESTION ID

S89-F1-Q1

TYPE

Short answer

CORRECT ANSWER

13.5

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, B = 60°, and a = 10. Find C, and find c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

3315

QUESTION ID

S89-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C = 80°; c ≈ 15.3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

True or false?

Knowing two angles and any side determines the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

3316

QUESTION ID

S89-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AAS/ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

A = 30°, B = 45°, a = 6. b =

A. 3√2 ≈ 4.2

B. 6√2 ≈ 8.5

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

3317

QUESTION ID

S89-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6√2 ≈ 8.5

GRADING NOTE 6 sin 45°/sin 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

Why must the third angle be found before using the Law of Sines for c in an ASA problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

3318

QUESTION ID

S89-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The law pairs each side with its opposite angle, so C is needed for c.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 2

Surveying with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 500 m apart sight a tree across a river. The angles at the observers (between the baseline and the tree) are 70° and 60°. How far is the tree from the observer at the 70° angle (nearest meter)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

3320

QUESTION ID
S89-F2-Q1

TYPE
Short answer

CORRECT ANSWER
565 m

GRADING NOTE Rounding to the nearest meter required. Third angle 50°; side opposite 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

3320

ANSWER KEY Reduced copy of the question page, with its answer

A surveyor measures a baseline of 200 m and angles of 52° and 71° to a peak. Find the distance from the 52° end to the peak (nearest meter).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

3321

QUESTION ID
S89-F2-Q2

TYPE
Short answer

CORRECT ANSWER
225 m

GRADING NOTE Rounding to the nearest meter required. Third angle 57°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

True or false?

Surveyors use a measured baseline and two angles to find distances they cannot walk.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

3322

QUESTION ID

S89-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Triangulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

In a surveying triangle, the side opposite the largest angle is

A. equal to the others

B. the shortest

C. the longest

D. the baseline always

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

3323

QUESTION ID

S89-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the longest

GRADING NOTE Law of Sines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

Explain why measuring angles is easier than measuring the distance across the river directly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

3324

QUESTION ID
S89-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The far side cannot be reached; angles can be measured from one bank.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Cosines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	The distance formula	**8.G.B.8**
BACKWARD 3	Sine and cosine in a right triangle	**HSG-SRT.C.6**
BACKWARD 4	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Use the Law of Cosines (SAS and SSS)	**HSG-SRT.D.10**
FORWARD 2	Distances between moving objects	**HSG-SRT.D.11**

SET 90 • HSG-SRT.D.10

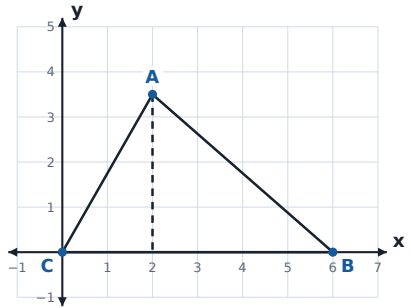
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Place triangle ABC with C at the origin and B at (a, 0), so A = (b cos C, b sin C). Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

3327

QUESTION ID

S90-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$c^2 = (b \cos C - a)^2 + (b \sin C)^2 = a^2 + b^2 - 2ab \cos C$

GRADING NOTE REQUIRED WORK: Use the distance formula for AB and the identity $\sin^2 C + \cos^2 C = 1$.

Any valid proof (paragraph, two-column, or flow). Essential: A = (b cos C, b sin C) because CA = b; $c^2 = (b \cos C - a)^2 + b^2 \sin^2 C$; expand: $b^2\cos^2 C - 2ab \cos C + a^2 + b^2 \sin^2 C$; $\sin^2 C + \cos^2 C = 1$ gives $a^2 + b^2 - 2ab \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In acute triangle ABC, the altitude from A meets BC at D. Let CD = x. Use the two right triangles to prove $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

3328

QUESTION ID

S90-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$h^2 = b^2 - x^2 = c^2 - (a - x)^2$, so $c^2 = a^2 + b^2 - 2ax = a^2 + b^2 - 2ab \cos C$.

GRADING NOTE REQUIRED WORK: Write h^2 twice with the Pythagorean Theorem, then use $x = b \cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: $BD = a - x$; $h^2 = b^2 - x^2$ and $h^2 = c^2 - (a - x)^2$; solve for c^2; substitute $x = b \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

Show that the Law of Cosines becomes the Pythagorean Theorem when C = 90°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

3329

QUESTION ID

S90-M-Q3

TYPE

Short answer

CORRECT ANSWER

cos 90° = 0, so $c^2 = a^2 + b^2$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If angle C is obtuse, then

A. $c = a + b$

B. $c^2 = a^2 + b^2$

C. $c^2 < a^2 + b^2$

D. $c^2 > a^2 + b^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

3330

QUESTION ID

S90-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $c^2 > a^2 + b^2$

GRADING NOTE cos C < 0, so −2ab cos C > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

True or false?

The Law of Cosines applies only to acute triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

3331

QUESTION ID

S90-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It holds for every triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Legs 5 and 12: hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

3333

QUESTION ID

S90-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b, leg x: the other leg squared?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

3334

QUESTION ID

S90-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$b^2 - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, the altitude from A to BC has length h and foot D, with CD = x and BC = a. In right triangle ABD, $h^2 + (a - x)^2 =$

A. a^2

B. c^2

C. x^2

D. b^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

3335

QUESTION ID

S90-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. c^2

GRADING NOTE Right triangle ABD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

True or false?

The Pythagorean Theorem applies to the right triangles made by an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

3336

QUESTION ID

S90-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Right angles at D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

True or false?

$a^2 + b^2 = c^2$ in every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

3337

QUESTION ID

S90-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 2

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (3, 4)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

3339

QUESTION ID

S90-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the squared distance from (p, q) to (a, 0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

3340

QUESTION ID

S90-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(p - a)^2 + q^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Squared distance from (1, 2) to (4, 6)?

A. 5

B. 25

C. 49

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

3341

QUESTION ID

S90-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

3342

QUESTION ID

S90-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

True or false?

Distance from (2, 0) to (5, 0) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

3343

QUESTION ID

S90-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 3

Sine and cosine in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b and angle C: adjacent side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

3345

QUESTION ID

S90-B3-Q1

TYPE

Short answer

CORRECT ANSWER

b cos C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-B3-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b and angle C: opposite side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

3346

QUESTION ID

S90-B3-Q2

TYPE

Short answer

CORRECT ANSWER

b sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-B3-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For a 3-4-5 triangle, $(\sin)^2 + (\cos)^2$ of an acute angle =

A. 25

B. 1

C. 12/25

D. 7/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

3347

QUESTION ID

S90-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE 9/25 + 16/25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^2 C + \cos^2 C = 1$ for acute angles C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

3348

QUESTION ID

S90-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

True or false?

cos C = opposite ÷ hypotenuse

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

3349

QUESTION ID
S90-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Adjacent ÷ hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 4

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(b \cos C - a)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

3351

QUESTION ID
S90-B4-Q1

TYPE
Short answer

CORRECT ANSWER
$b^2\cos^2 C - 2ab\cos C + a^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a - x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

3352

QUESTION ID
S90-B4-Q2

TYPE
Short answer

CORRECT ANSWER
$a^2 - 2ax + x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(p - q)^2 =$

A. $p^2 - 2pq + q^2$

B. $p^2 - pq + q^2$

C. $p^2 - q^2$

D. $p^2 + q^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

3353

QUESTION ID
S90-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $p^2 - 2pq + q^2$

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - x)^2 - x^2 = a^2 - 2ax$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

3354

QUESTION ID
S90-B4-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE x^2 cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - x)^2 = a^2 - x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

3355

QUESTION ID
S90-B4-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing −2ax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 1

Use the Law of Cosines (SAS and SSS) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, a = 7, b = 9, and C = 50°. Find c to the nearest tenth.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

3357

QUESTION ID

S90-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.0

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 7.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

3358

QUESTION ID

S90-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form. cos = (25 + 64 − 49)/80 = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

True or false?

The Law of Cosines can find an angle when all three sides are known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

3359

QUESTION ID

S90-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Sides 3, 4, 6. The angle opposite 6 is

A. obtuse

B. acute

C. right

D. impossible to tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

3360

QUESTION ID

S90-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. obtuse

GRADING NOTE $36 > 9 + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the largest angle of the triangle with sides 4, 5, and 6, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

3361

QUESTION ID

S90-F1-Q5

TYPE

Short answer

CORRECT ANSWER

82.8°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 2

Distances between moving objects • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Two ships leave port at the same time on courses 70° apart. One travels 20 km and the other 30 km. How far apart are they, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

3363

QUESTION ID

S90-F2-Q1

TYPE

Short answer

CORRECT ANSWER

29.8 km

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A golfer's ball is 150 yd from the tee; the hole is 180 yd from the tee; the angle between the two directions is 12°. How far is the ball from the hole (nearest yard)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

3364

QUESTION ID
S90-F2-Q2

TYPE
Short answer

CORRECT ANSWER
46 yd

GRADING NOTE Rounding to the nearest yard required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

True or false?

The Law of Cosines can be used only when the angle between the two known sides is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

3365

QUESTION ID

S90-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It works for any included angle; with 90° it becomes the Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two hikers walk 3 km and 5 km from camp in directions 60° apart. Distance between them?

A. 8 km

B. √34 ≈ 5.8 km

C. 2 km

D. √19 ≈ 4.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

3366

QUESTION ID

S90-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √19 ≈ 4.4 km

GRADING NOTE 9 + 25 − 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two ships leave port on courses 70° apart. Why can't the Pythagorean Theorem alone give the distance between them?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

3367

QUESTION ID
S90-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The angle between their courses is not 90°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Sines, including the ambiguous case

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Sines	**HSG-SRT.D.10**
BACKWARD 2	Inverse sine	**HSG-SRT.C.8**
BACKWARD 3	Triangle angle sum	**HSG-CO.C.10**
BACKWARD 4	Sine of supplementary angles	**HSG-SRT.D.9**
FORWARD 1	Solve both triangles in the ambiguous case	**HSG-SRT.D.11**
FORWARD 2	Bearings and navigation with the Law of Sines	**HSG-MG.A.1**

SET 91 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Points A and B are 120 m apart on one bank of a river. A tree C is on the other bank. ∠A = 72° and ∠B = 63°. Find AC to the nearest tenth of a meter.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

3370

QUESTION ID

S91-M-Q1

TYPE

Short answer

CORRECT ANSWER

151.2 m

GRADING NOTE Rounding to 0.1 required. ∠C = 45°; AC is opposite ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 30°, a = 6, and b = 10. How many triangles are possible? Find all possible measures of B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

3371

QUESTION ID

S91-M-Q2

TYPE

Short answer

CORRECT ANSWER

Two: B ≈ 56.4° or B ≈ 123.6°

GRADING NOTE Form required: the number and both angles. sin B = 10 sin 30°/6 ≈ 0.833.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, a = 12, and b = 10. How many triangles are possible? Find B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

3372

QUESTION ID

S91-M-Q3

TYPE

Short answer

CORRECT ANSWER

One: B ≈ 32.4°

GRADING NOTE Form required: the number and the angle. The supplement 147.6° is too large (with A = 40° the sum exceeds 180°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 50°, a = 5, and b = 10. How many triangles are possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

3373

QUESTION ID

S91-M-Q4

TYPE

Short answer

CORRECT ANSWER

None: sin B = 10 sin 50°/5 ≈ 1.53 > 1.

GRADING NOTE Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

A is acute and b sin A < a < b. How many triangles have these measurements?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

3374

QUESTION ID

S91-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Two

GRADING NOTE Side a can swing to two positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 1

The Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Write the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

3376

QUESTION ID

S91-B1-Q1

TYPE

Short answer

CORRECT ANSWER

a/sin A = b/sin B = c/sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 90°, a = 5. b =

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

3377

QUESTION ID

S91-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving a/sin A = b/sin B for sin B gives

A. b sin A/a

B. a/(b sin A)

C. sin A

D. a sin A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

3378

QUESTION ID

S91-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b sin A/a

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

True or false?

Each side in the Law of Sines is paired with its opposite angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

3379

QUESTION ID

S91-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a/\sin B = b/\sin A$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

3380

QUESTION ID

S91-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Pairs are opposite.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 2

Inverse sine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin B = 0.5, B acute. B =

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

3382

QUESTION ID

S91-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin B = 0.8. Find the acute B to the nearest tenth.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

3383

QUESTION ID

S91-B2-Q2

TYPE

Short answer

CORRECT ANSWER

53.1°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin^{-1}(1) =$

A. 0°

B. 180°

C. 45°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

3384

QUESTION ID

S91-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90°

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The calculator's $\sin^{-1}$ always gives an angle between $-90°$ and $90°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

3385

QUESTION ID

S91-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^{-1}(1.2)$ is an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

3386

QUESTION ID

S91-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Sine cannot exceed 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 3

Triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Two angles are 72° and 63°. Third angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

3388

QUESTION ID

S91-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A triangle cannot have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

3389

QUESTION ID

S91-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 40° and B = 147.6°. Is this a triangle?

A. Only if C is 0°

B. No: the sum exceeds 180°

C. Yes

D. Only if right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

3390

QUESTION ID

S91-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. No: the sum exceeds 180°

GRADING NOTE 187.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 123.6°. Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

3391

QUESTION ID

S91-B3-Q4

TYPE

Short answer

CORRECT ANSWER

26.4°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angles of a triangle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

3392

QUESTION ID

S91-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 4

Sine of supplementary angles • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 150° = \sin 30°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

3394

QUESTION ID

S91-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sin(180° - x) = \sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name an obtuse angle with the same sine as 56.4°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

3395

QUESTION ID

S91-B4-Q2

TYPE

Short answer

CORRECT ANSWER

123.6°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 120° equals

A. cos 120°

B. sin 60°

C. sin 30°

D. −sin 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

3396

QUESTION ID

S91-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. sin 60°

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why might sin B = 0.83 give two possible triangles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

3397

QUESTION ID

S91-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Both B and 180° − B have that sine.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

True or false?

An obtuse angle has a negative sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

3398

QUESTION ID

S91-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 1

Solve both triangles in the ambiguous case • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, a = 6, b = 10. For the triangle with acute B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

3400

QUESTION ID

S91-F1-Q1

TYPE

Short answer

CORRECT ANSWER

C ≈ 93.6°; c ≈ 12.0

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10, so B ≈ 56.4° or B ≈ 123.6°. For the triangle with obtuse B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

3401

QUESTION ID

S91-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C ≈ 26.4°; c ≈ 5.3

GRADING NOTE Form required: both. B ≈ 123.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

True or false?

In the ambiguous case the two triangles have different third sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

3402

QUESTION ID
S91-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

A = 30°, b = 10, and a = 5. How many triangles?

A. Three

B. Two

C. One (a right triangle)

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

3403

QUESTION ID

S91-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. One (a right triangle)

GRADING NOTE a = b sin A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 120°, a = 8, b = 10. How many triangles? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

3404

QUESTION ID

S91-F1-Q5

TYPE

Short answer

CORRECT ANSWER

None: with A obtuse, a must be the longest side, but a < b.

GRADING NOTE Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 2

Bearings and navigation with the Law of Sines • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A lighthouse is spotted from a ship at an angle of 35° to its course. After the ship sails 8 km farther, the angle is 70°. How far is the ship from the lighthouse at the second sighting (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3406

QUESTION ID

S91-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8.0 km

GRADING NOTE Rounding to 0.1 required. The triangle has angles 35°, 110°, 35°, so it is isosceles: 8.0 km.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3406

ANSWER KEY Reduced copy of the question page, with its answer

A forest ranger at A sees a fire at an angle of 48° from the line to a second tower B, 10 km away. From B, the angle is 61°. How far is the fire from A (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

3407

QUESTION ID

S91-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9.3 km

GRADING NOTE Rounding to 0.1 required. Third angle 71°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

True or false?

Bearings and sightings are modeled as triangles with a known side and two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

3408

QUESTION ID

S91-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A ship sights a lighthouse at 35° to its course; after sailing 8 km farther along the course, the angle is 70°. In the triangle formed by the two ship positions and the lighthouse, the angle at the lighthouse is

A. 75°

B. 110°

C. 70°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

3409

QUESTION ID

S91-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 35°

GRADING NOTE 70° − 35° (exterior angle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name one reason a navigation model using a flat triangle might be inaccurate over very long distances.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

3410

QUESTION ID
S91-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Earth's surface is curved.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

3410

SET 92 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Cosines, including resultant forces

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Cosines	**HSG-SRT.D.10**
BACKWARD 2	Inverse cosine	**HSG-SRT.C.8**
BACKWARD 3	Angles in a parallelogram	**HSG-CO.C.11**
FORWARD 1	Solve a triangle with both laws	**HSG-SRT.D.11**
FORWARD 2	Velocity vectors: airspeed and wind	**HSN-VM.A.3**

SET 92 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

To find the length AB across a lake, a surveyor at C measures CA = 350 m, CB = 420 m, and ∠C = 68°. Find AB to the nearest tenth of a meter.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

3413

QUESTION ID

S92-M-Q1

TYPE

Short answer

CORRECT ANSWER

434.5 m

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

3413

ANSWER KEY Reduced copy of the question page, with its answer

Two forces of 40 N and 60 N act on an object with an angle of 50° between them. Find the magnitude of the resultant force to the nearest tenth.

Resultant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

3414

QUESTION ID

S92-M-Q2

TYPE

Short answer

CORRECT ANSWER

91.0 N

GRADING NOTE Rounding to 0.1 required. In the force parallelogram the angle opposite the resultant is 130°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

3414

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 8, 10, and 13. Find its largest angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

3415

QUESTION ID

S92-M-Q3

TYPE

Short answer

CORRECT ANSWER

91.8°

GRADING NOTE Rounding to 0.1 required. Opposite 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane flies 200 km, turns 25° to the right, and flies 150 km. How far is it from its start, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

3416

QUESTION ID

S92-M-Q4

TYPE

Short answer

CORRECT ANSWER

341.9 km

GRADING NOTE Rounding to 0.1 required. The angle between the legs inside the triangle is 155°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

Which law should be used first to solve a triangle given all three sides?

A. The Law of Sines

B. The Pythagorean Theorem

C. The angle sum

D. The Law of Cosines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

3417

QUESTION ID

S92-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The Law of Cosines

GRADING NOTE SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 1

The Law of Cosines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Write the Law of Cosines for side c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

3419

QUESTION ID

S92-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$c^2 = a^2 + b^2 - 2ab \cos C$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

a = 3, b = 4, C = 90°: c?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

3420

QUESTION ID

S92-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

3420

ANSWER KEY Reduced copy of the question page, with its answer

Solved for cos C:

A. $(a^2 + b^2 - c^2)/(2ab)$

B. $(c^2 - a^2 - b^2)/(2ab)$

C. $2ab/(a^2 + b^2 - c^2)$

D. $(a^2 + b^2 + c^2)/(2ab)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

3421

QUESTION ID

S92-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

$(a^2 + b^2 - c^2)/(2ab)$

GRADING NOTE Rearrange.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

True or false?

The angle in the formula is the one opposite the side being found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

3422

QUESTION ID

S92-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE C opposite c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

True or false?

$c^2 = a^2 + b^2 + 2ab \cos C$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

3423

QUESTION ID

S92-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Minus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 2

Inverse cosine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

cos C = 0.5. C =

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

3425

QUESTION ID

S92-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos C = −0.03125. C to the nearest tenth?

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

3426

QUESTION ID

S92-B2-Q2

TYPE

Short answer

CORRECT ANSWER

91.8°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A negative cosine means the angle is

A. acute

B. right

C. obtuse

D. zero

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

3427

QUESTION ID

S92-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. obtuse

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

True or false?

$\cos^{-1}$ gives angles from 0° to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

3428

QUESTION ID

S92-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos^{-1}(0) = 0°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

3429

QUESTION ID

S92-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 3

Angles in a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

3431

QUESTION ID

S92-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

One angle of a parallelogram is 50°. The angle next to it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

3432

QUESTION ID

S92-B3-Q2

TYPE

Short answer

CORRECT ANSWER

130°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Forces at 50° form a parallelogram; the angle opposite the resultant in the triangle is

A. 130°

B. 25°

C. 50°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

3433

QUESTION ID

S92-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 130°

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane turns 25°. The angle inside the triangle at the turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

3434

QUESTION ID

S92-B3-Q4

TYPE

Short answer

CORRECT ANSWER

155°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

True or false?

The angle between two forces is the angle used in the Law of Cosines for the resultant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3435

QUESTION ID
S92-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Use its supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 1

Solve a triangle with both laws • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, A = 40°. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

3437

QUESTION ID
S92-F1-Q1

TYPE
Short answer

CORRECT ANSWER
9.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, and A = 40°. Find B to the nearest tenth of a degree (use the Law of Sines or Cosines).

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

3438

QUESTION ID

S92-F1-Q2

TYPE

Short answer

CORRECT ANSWER

53.0°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

True or false?

After the Law of Cosines, it is safest to find the smaller remaining angle with the Law of Sines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

3439

QUESTION ID

S92-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Avoids the ambiguous obtuse case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

Given two sides and the included angle, use first

A. $\sin^{-1}$

B. the Law of Sines

C. the Law of Cosines

D. the angle sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

3440

QUESTION ID

S92-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the Law of Cosines

GRADING NOTE SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use Heron's formula or ½ab sin C to find the area of the triangle with sides 8, 10, 13 (nearest tenth).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

3441

QUESTION ID
S92-F1-Q5

TYPE
Short answer

CORRECT ANSWER
40.0

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 2

Velocity vectors: airspeed and wind • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A plane flies due east at 300 km/h; a wind blows due north at 50 km/h. Find its ground speed (nearest tenth) and direction (degrees north of east, nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

3443

QUESTION ID
S92-F2-Q1

TYPE
Short answer

CORRECT ANSWER
304.1 km/h at 9.5° north of east

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A boat heads across a river at 8 km/h; the current is 3 km/h at 90° to its heading. Find its actual speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

3444

QUESTION ID
S92-F2-Q2

TYPE
Short answer

CORRECT ANSWER
8.5 km/h

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

3444

True or false?

The resultant of two velocities is always found by adding their speeds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

3445

QUESTION ID

S92-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when they point the same way; otherwise use the parallelogram rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

3445

ANSWER KEY Reduced copy of the question page, with its answer

Velocities of 10 and 10 with 120° between them have a resultant of

A. 0

B. 17.3

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

3446

QUESTION ID

S92-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE Law of Cosines with 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane's velocity is 400 km/h; a 60 km/h wind blows at 45° to its heading. Find the ground speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

3447

QUESTION ID

S92-F2-Q5

TYPE

Short answer

CORRECT ANSWER

444.5 km/h

GRADING NOTE Rounding to 0.1 required. Angle in the triangle 135°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

SET 93 • HSG-C.A.4 • ADVANCED EXTENSION (+)

Construct a tangent line from a point outside a circle

MAIN	Geometry main question	**HSG-C.A.4**
BACKWARD 1	Construct the midpoint of a segment	**HSG-CO.D.12**
BACKWARD 2	Inscribed angles on a diameter	**HSG-C.A.2**
BACKWARD 3	Tangent is perpendicular to the radius	**HSG-C.A.2**
FORWARD 1	Prove that the two tangent segments are congruent and compute…	**HSG-C.A.4**
FORWARD 2	Lines of sight to the horizon	**HSG-MG.A.1**

SET 93 • HSG-C.A.4

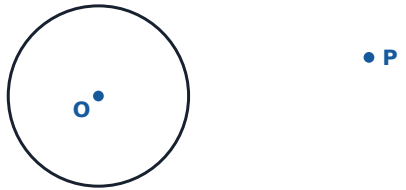
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a line through P tangent to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

3450

QUESTION ID

S93-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw OP; construct its midpoint M (perpendicular bisector); draw the circle with center M through O; it meets circle O at T_1 and T_2; draw PT_1 (and PT_2).

GRADING NOTE Grade the construction marks, not only the final figure: segment OP, the perpendicular bisector arcs locating M, the circle on diameter OP, and the tangent through a crossing point T.

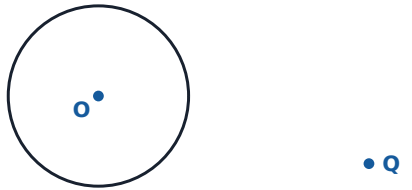
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct BOTH tangent lines from Q to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

3451

QUESTION ID

S93-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same construction; both intersection points T_1 and T_2 give the two tangents QT_1 and QT_2.

GRADING NOTE Grade the construction marks, not only the final figure: all construction marks and both tangents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the steps for constructing a tangent from an external point P to circle O.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

3452

QUESTION ID

S93-M-Q3

TYPE

Short answer

CORRECT ANSWER

1. Draw OP. 2. Bisect OP to find its midpoint M. 3. Draw the circle with center M and radius MO. 4. Label a point T where it meets circle O. 5. Draw line PT.

GRADING NOTE Any equivalent steps.

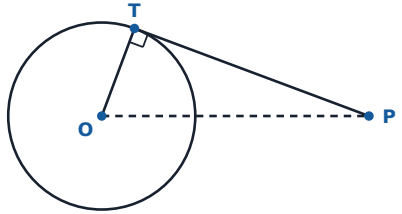
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why line PT in the construction is tangent to circle O.

Work space



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

3453

QUESTION ID

S93-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

T is on the circle with diameter OP, so ∠OTP is inscribed in a semicircle and equals 90°; a line perpendicular to a radius at its endpoint on the circle is tangent.

GRADING NOTE

REQUIRED WORK: Use the circle on diameter OP and the tangent-radius theorem.

Any valid explanation. Essential: ∠OTP = 90° (inscribed angle on a diameter); PT ⊥ OT at T on circle O; converse of the tangent-radius theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

The two tangent segments from P to circle O are

A. perpendicular

B. congruent

C. parallel

D. always equal to the radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

3454

QUESTION ID

S93-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. congruent

GRADING NOTE HL congruence of triangles OTP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 1

Construct the midpoint of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Which construction finds the midpoint of OP?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

3456

QUESTION ID

S93-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of OP

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector crosses the segment at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

3457

QUESTION ID

S93-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

The perpendicular bisector construction uses arcs

A. of any radius

B. centered at the midpoint

C. of equal radius from both endpoints

D. from one endpoint only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

3458

QUESTION ID

S93-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. of equal radius from both endpoints

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct a tangent from a point P outside circle O, why is the midpoint of OP needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

3459

QUESTION ID

S93-B1-Q4

TYPE

Short answer

CORRECT ANSWER

It is the center of the circle with diameter OP.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

True or false?

The midpoint can be found by eye in a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

3460

QUESTION ID

S93-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It must be constructed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 2

Inscribed angles on a diameter • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

An angle inscribed in a semicircle is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

3462

QUESTION ID

S93-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point T is on the circle with diameter OP. m∠OTP = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

3463

QUESTION ID

S93-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The intercepted arc of an angle inscribed in a semicircle is

A. 180°

B. 360°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

3464

QUESTION ID

S93-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 180°

GRADING NOTE Half of 180 is 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct a tangent from a point P outside circle O, why is a circle with diameter OP drawn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

3465

QUESTION ID

S93-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Points on it make a right angle with O and P.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

True or false?

An inscribed angle on a diameter can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

3466

QUESTION ID

S93-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 3

Tangent is perpendicular to the radius • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

3468

QUESTION ID

S93-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tangent meets the radius at what angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

3469

QUESTION ID

S93-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

If PT ⊥ OT and T is on circle O, then PT is

A. a secant

B. a diameter

C. a chord

D. tangent to the circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

3470

QUESTION ID

S93-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. tangent to the circle

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

OT = 5 and OP = 13 with PT tangent at T. PT = ?

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

3471

QUESTION ID

S93-B3-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

True or false?

A tangent passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

3472

QUESTION ID

S93-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches at one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 1

Prove that the two tangent segments are congruent and compute lengths • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

PA and PB are tangent to circle O at A and B. Prove that PA = PB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

3474

QUESTION ID

S93-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△OAP ≅ △OBP by HL, so PA = PB.

GRADING NOTE REQUIRED WORK: Use the right triangles OAP and OBP.

Any valid proof (paragraph, two-column, or flow). Essential: right angles at A and B (tangent ⊥ radius); OA = OB (radii); OP common hypotenuse; HL; CPCTC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 5 and P is 13 from its center. Find the length of each tangent segment from P.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

3475

QUESTION ID

S93-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

True or false?

OP bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

3476

QUESTION ID

S93-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P make a 60° angle. The angle AOB is

A. 30°

B. 120°

C. 60°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

3477

QUESTION ID

S93-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 120°

GRADING NOTE Quadrilateral with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Radius 6, tangent segment 8. Distance from P to the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

3478

QUESTION ID

S93-F1-Q5

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 2

Lines of sight to the horizon • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere of radius 6,371 km. A person's eyes are 2 m (0.002 km) above the sea. How far is the horizon (tangent distance), to the nearest tenth of a km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

3480

QUESTION ID

S93-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5.0 km

GRADING NOTE Rounding to 0.1 required. $d^2 = (6371.002)^2 - 6371^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From the top of a 100 m tower (0.1 km), how far is the horizon, to the nearest km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

3481

QUESTION ID

S93-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36 km

GRADING NOTE Rounding to the nearest km required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The line of sight to the horizon passes through Earth's center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3482

QUESTION ID

S93-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is tangent, perpendicular to the radius at the horizon point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Raising the eye height by a factor of 4 multiplies the horizon distance by about

A. 16

B. 8

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

3483

QUESTION ID

S93-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $d \approx \sqrt{(2Rh)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

3483

ANSWER KEY Reduced copy of the question page, with its answer

In the model of the distance to the horizon from a point above Earth, why is the right angle of the triangle at the horizon point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

3484

QUESTION ID

S93-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The tangent line of sight is perpendicular to the radius at the point of tangency.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of an ellipse from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive a circle equation	**HSG-GPE.A.1**
BACKWARD 3	Expand squares	**HSA-APR.A.1**
FORWARD 1	Read foci and axes from an ellipse equation	**HSG-GPE.A.3**
FORWARD 2	Design with the reflective property of ellipses	**HSG-MG.A.3**

SET 94 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An ellipse is the set of points whose distances to the foci $(-3, 0)$ and $(3, 0)$ add to 10. Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

3487

QUESTION ID

S94-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$$x^2/25 + y^2/16 = 1$$

GRADING NOTE REQUIRED WORK: Write the distance sum, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + 3)^2 + y^2)} + \sqrt{((x - 3)^2 + y^2)} = 10$; isolate and square; simplify to $3x - 25 = -5\sqrt{((x - 3)^2 + y^2)}$ (or equivalent); square again; $16x^2 + 25y^2 = 400$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the ellipse with foci (0, −4) and (0, 4) whose distances to the foci add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

3488

QUESTION ID

S94-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2/9 + y^2/25 = 1$

GRADING NOTE Form required: an equation. a = 5 (vertical), c = 4, $b^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

Write the equation of the ellipse with foci (±5, 0) and vertices (±13, 0).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

3489

QUESTION ID

S94-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^2/169 + y^2/144 = 1$

GRADING NOTE Form required: an equation. $b^2 = 169 - 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

Write the equation of the ellipse with foci (1, 2) and (7, 2) whose distances to the foci add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

3490

QUESTION ID

S94-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 4)^2/25 + (y - 2)^2/16 = 1$

GRADING NOTE Form required: an equation. Center (4, 2), c = 3, a = 5, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

For any point on $x^2/36 + y^2/20 = 1$, the sum of its distances to the foci is

A. 12

B. $2\sqrt{20}$

C. 36

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

3491

QUESTION ID

S94-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 12

GRADING NOTE 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (x, y) to (3, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

3493

QUESTION ID

S94-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{((x - 3)^2 + y^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 4) to (3, 0)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

3494

QUESTION ID

S94-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (5, 0) to (−3, 0)?

A. √34

B. 64

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

3495

QUESTION ID

S94-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

True or false?

The distance formula has a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

3496

QUESTION ID

S94-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(a^2 + b^2)} = a + b$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

3497

QUESTION ID

S94-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 2

Derive a circle equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Equation of the circle with center (0, 0) and radius 5?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

3499

QUESTION ID

S94-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-B2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

True or false?

A circle is an ellipse whose two foci coincide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

3500

QUESTION ID

S94-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Special case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2/25 + y^2/25 = 1$ is

A. an ellipse with different axes

B. a parabola

C. a circle of radius 5

D. a hyperbola

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

3501

QUESTION ID

S94-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a circle of radius 5

GRADING NOTE Equal denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is the set of points at a fixed distance from

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

3502

QUESTION ID

S94-B2-Q4

TYPE

Short answer

CORRECT ANSWER

one point (the center)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An ellipse has one focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

3503

QUESTION ID

S94-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 3

Expand squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 3)^2 - (x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

3505

QUESTION ID
S94-B3-Q1

TYPE
Short answer

CORRECT ANSWER
12x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

3506

QUESTION ID
S94-B3-Q2

TYPE
Short answer

CORRECT ANSWER
$x^2 - 6x + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(3x - 25)^2 =$

A. $9x^2 + 625$

B. $9x^2 - 75x + 625$

C. $3x^2 - 150x + 625$

D. $9x^2 - 150x + 625$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

3507

QUESTION ID
S94-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $9x^2 - 150x + 625$

GRADING NOTE Middle term $2 \cdot 3x \cdot 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - b)^2 = a^2 - 2ab + b^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

3508

QUESTION ID
S94-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

3509

QUESTION ID
S94-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 1

Read foci and axes from an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the foci of $x^2/25 + y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

3511

QUESTION ID

S94-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(±4, 0)

GRADING NOTE Form required: both points. $c^2 = 25 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertices and foci of $x^2/16 + y^2/25 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

3512

QUESTION ID

S94-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertices (0, ±5); foci (0, ±3)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

True or false?

For an ellipse, $c^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

3513

QUESTION ID

S94-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

Which ellipse is closest to a circle?

A. $x^2/100 + y^2/9 = 1$

B. $x^2/25 + y^2/4 = 1$

C. $x^2/25 + y^2/24 = 1$

D. $x^2/25 + y^2/1 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

3514

QUESTION ID

S94-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2/25 + y^2/24 = 1$

GRADING NOTE Nearly equal denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the major axis of $x^2/49 + y^2/36 = 1$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

3515

QUESTION ID

S94-F1-Q5

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 2

Design with the reflective property of ellipses • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A whispering gallery is an ellipse 40 m long and 16 m wide. How far from the center should two people stand (at the foci), to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

3517

QUESTION ID

S94-F2-Q1

TYPE

Short answer

CORRECT ANSWER

18.3 m

GRADING NOTE Rounding to 0.1 required. a = 20, b = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A medical device places a kidney stone at one focus of an ellipse $x^2/100 + y^2/64 = 1$ (cm). Where is the emitter placed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

3518

QUESTION ID

S94-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At the other focus, 6 cm from the center: (−6, 0) if the stone is at (6, 0).

GRADING NOTE Any equivalent answer. c = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

True or false?

Sound from one focus of an elliptical room reflects back to the same focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

3519

QUESTION ID

S94-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It reflects to the other focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

A garden ellipse is drawn with a 10 m rope tied to stakes 6 m apart. Its width (minor axis) is

A. 8 m

B. 10 m

C. 4 m

D. 6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

3520

QUESTION ID

S94-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8 m

GRADING NOTE a = 5, c = 3, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

3520

Explain how the string-and-stakes method uses the definition of an ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

3521

QUESTION ID

S94-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The string length is the constant sum of the distances to the two stakes (foci).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of a hyperbola from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive an ellipse equation	**HSG-GPE.A.3**
BACKWARD 3	Absolute value as distance	**6.NS.C.7**
FORWARD 1	Read foci and asymptotes from a hyperbola equation	**HSG-GPE.A.3**
FORWARD 2	Locate a position from a time difference	**HSG-MG.A.3**

SET 95 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A hyperbola is the set of points whose distances to the foci (−5, 0) and (5, 0) differ by 6 (in absolute value). Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

3524

QUESTION ID

S95-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2/9 - y^2/16 = 1$

GRADING NOTE REQUIRED WORK: Write the distance difference, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{((x + 5)^2 + y^2)} - \sqrt{((x - 5)^2 + y^2)}| = 6$; isolate and square; simplify to $5x - 9 = \pm 3\sqrt{((x - 5)^2 + y^2)}$ (or equivalent); square again; $16x^2 - 9y^2 = 144$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

Write the equation of the hyperbola with foci (0, ±13) whose distances to the foci differ by 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

3525

QUESTION ID

S95-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y^2/25 - x^2/144 = 1$

GRADING NOTE Form required: an equation. $a = 5$, $c = 13$, $b^2 = 144$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

Write the equation of the hyperbola with foci (±10, 0) and vertices (±6, 0).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

3526

QUESTION ID

S95-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^2/36 - y^2/64 = 1$

GRADING NOTE Form required: an equation. $b^2 = 100 - 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the asymptotes of $x^2/9 - y^2/16 = 1$.

Asymptotes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

3527

QUESTION ID

S95-M-Q4

TYPE

Short answer

CORRECT ANSWER

$y = \pm(4/3)x$

GRADING NOTE Form required: both lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

For any point on $x^2/16 - y^2/9 = 1$, the difference of its distances to the foci is

A. 8

B. 7

C. 10

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

3528

QUESTION ID

S95-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 8

GRADING NOTE 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (x, y) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

3530

QUESTION ID

S95-B1-Q1

TYPE

Short answer

CORRECT ANSWER

√((x + 5)² + y²)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (3, 0) to (5, 0) minus distance from (3, 0) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

3531

QUESTION ID

S95-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−6 (difference 6 in absolute value)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (6, 0) to (−10, 0)?

A. 4

B. 60

C. 16

D. −16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

3532

QUESTION ID

S95-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

True or false?

Distances are never negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

3533

QUESTION ID

S95-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

True or false?

Distance from (0, 13) to (0, −13) is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

3534

QUESTION ID

S95-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 2

Derive an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

True or false?

An ellipse uses a constant SUM of distances to the foci.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

3536

QUESTION ID

S95-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Ellipse with foci (±3, 0) and sum 10: equation?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

3537

QUESTION ID

S95-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2/25 + y^2/16 = 1$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For an ellipse, $b^2 =$

A. ac

B. $a^2 - c^2$

C. $a^2 + c^2$

D. $c^2 - a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

3538

QUESTION ID

S95-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a^2 - c^2$

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What changes in the definition to get a hyperbola?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

3539

QUESTION ID

S95-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The difference (not the sum) of the distances is constant.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

True or false?

An ellipse equation has a minus sign between the squared terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

3540

QUESTION ID

S95-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Plus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 3

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

ANSWER KEY Reduced copy of the question page, with its answer

$|4 - 10| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

3542

QUESTION ID

S95-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

3542

ANSWER KEY Reduced copy of the question page, with its answer

$|10 - 4| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

3543

QUESTION ID

S95-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

|a − b| means

A. the size of the difference

B. a + b

C. a negative number

D. always a − b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

3544

QUESTION ID

S95-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the size of the difference

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

True or false?

$|x| = 6$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

3545

QUESTION ID

S95-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ±6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

True or false?

$|-3| = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

3546

QUESTION ID

S95-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 1

Read foci and asymptotes from a hyperbola equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the foci of $x^2/16 - y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3548

QUESTION ID

S95-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(±5, 0)

GRADING NOTE Form required: both points. $c^2 = 16 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3548

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertices and asymptotes of $y^2/4 - x^2/9 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

3549

QUESTION ID

S95-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertices (0, ±2); asymptotes y = ±(2/3)x

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For a hyperbola, $c^2 = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

3550

QUESTION ID

S95-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation is a hyperbola?

A. $x^2/4 + y^2/9 = 1$

B. $y = x^2/4$

C. $x^2/4 - y^2/9 = 1$

D. $x^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

3551

QUESTION ID

S95-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2/4 - y^2/9 = 1$

GRADING NOTE Minus sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

Explain how to tell an ellipse from a hyperbola by its equation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

3552

QUESTION ID
S95-F1-Q5

TYPE
Short answer

CORRECT ANSWER
An ellipse has a plus sign between the squared terms; a hyperbola has a minus sign.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 2

Locate a position from a time difference • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Two radio stations are 100 km apart at (±50, 0). A ship receives signals showing it is 60 km closer to one station than the other. Write the equation of the hyperbola it lies on.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

3554

QUESTION ID

S95-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2/900 - y^2/1600 = 1$

GRADING NOTE Form required: an equation. $2a = 60$, $c = 50$, $b^2 = 2500 - 900$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sound travels 0.34 km/s. Two microphones 2 km apart hear a sound 3 s apart. What is the difference in distances to the source?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

3555

QUESTION ID

S95-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.02 km

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

True or false?

A single time difference places the source on one branch of a hyperbola, not at a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3556

QUESTION ID

S95-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A second pair of stations is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3556

Radio stations are at (−50, 0) and (50, 0) (km). A ship on the x-axis, to the right of the origin, is 60 km closer to the right station than to the left one. Its position is

A. (30, 0)

B. (40, 0)

C. (50, 0)

D. (60, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

3557

QUESTION ID

S95-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (30, 0)

GRADING NOTE Vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

Why do navigation systems use at least two pairs of stations?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

3558

QUESTION ID
S95-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Each pair gives a hyperbola; the intersection pins down the position.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2 • ADVANCED EXTENSION (+)

Use Cavalieri's principle to justify the volume of a sphere

MAIN	Geometry main question	**HSG-GMD.A.2**
BACKWARD 1	Cavalieri's principle for prisms and cylinders	**HSG-GMD.A.1**
BACKWARD 2	Cross-section radius with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Volumes of cylinders and cones	**HSG-GMD.A.3**
BACKWARD 4	Area of a ring	**7.G.B.4**
FORWARD 1	Use Cavalieri's principle for other solids	**HSG-GMD.A.2**
FORWARD 2	Model a dome as a hemispherical shell	**HSG-MG.A.2**

SET 96 • HSG-GMD.A.2

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius r and a cylinder of radius r and height r with a cone removed (as shown) sit on the same plane. Use Cavalieri's principle to show that the hemisphere's volume is ⅔πr³, and conclude that a sphere has volume 4/3 πr³.

Work space



hemisphere, radius r cylinder minus cone

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

3561

QUESTION ID

S96-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Both cross-sections have area π(r² − y²), so the volumes are equal: πr³ − ⅓πr³ = ⅔πr³; the sphere is two hemispheres: 4/3 πr³.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas at height y.

Any valid explanation. Essential: hemisphere slice radius √(r² − y²) (Pythagorean), area π(r² − y²); the other slice is a ring with outer radius r and inner radius y (the cone radius at height y), area πr² − πy²; equal areas at every height and equal heights; Cavalieri; volume cylinder minus cone = ⅔πr³; double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Let r = 5 and y = 3. Find the cross-section areas of the hemisphere and of the cylinder-minus-cone at height y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

3562

QUESTION ID

S96-M-Q2

TYPE

Short answer

CORRECT ANSWER

Both 16π

GRADING NOTE Form required: both. 25π − 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Let r = 6. Find the volume of the cylinder-minus-cone and compare it with $\frac{2}{3}\pi r^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

3563

QUESTION ID

S96-M-Q3

TYPE

Short answer

CORRECT ANSWER

$216\pi - 72\pi = 144\pi = \frac{2}{3}\pi(216)$

GRADING NOTE Form required: the computation and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why a stack of coins pushed into a slanted stack keeps its volume, using Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

3564

QUESTION ID
S96-M-Q4

TYPE
Short answer

CORRECT ANSWER
Each coin (cross-section) has the same area at every height, so the volumes are equal.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

Cavalieri's principle requires

A. equal surface areas

B. congruent bases only

C. the same shape at every level

D. equal heights and equal cross-section areas at every level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

3565

QUESTION ID

S96-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. equal heights and equal cross-section areas at every level

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 1

Cavalieri's principle for prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Solids with equal heights and equal cross-section areas at every level have equal volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

3567

QUESTION ID

S96-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Principle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cylinder has radius 2 and height 5. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

3568

QUESTION ID

S96-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

A prism and a cylinder both have base area 30 and height 4. Volumes?

A. Cannot tell

B. 30 and 4

C. 120 and 40

D. Both 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

3569

QUESTION ID

S96-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Both 120

GRADING NOTE Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

State what must be equal for Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

3570

QUESTION ID

S96-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The heights and the cross-section areas at every level.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Cavalieri's principle needs the cross-sections to be the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

3571

QUESTION ID
S96-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only equal areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 2

Cross-section radius with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius 5 is sliced 3 units from its center. Radius of the slice?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

3573

QUESTION ID

S96-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Radius r, height y: slice radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

3574

QUESTION ID

S96-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(r^2 - y^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

Radius 10, height 6: slice radius

A. 4

B. 8

C. 16

D. √136

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

3575

QUESTION ID

S96-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

GRADING NOTE 100 − 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

True or false?

The slice radius, the height, and the sphere radius form a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

3576

QUESTION ID

S96-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Right angle at the slice center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

True or false?

Slices of a hemisphere get bigger as y increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

3577

QUESTION ID

S96-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 3

Volumes of cylinders and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

3579

QUESTION ID

S96-B3-Q1

TYPE

Short answer

CORRECT ANSWER

216π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cone r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

3580

QUESTION ID

S96-B3-Q2

TYPE

Short answer

CORRECT ANSWER

72π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder minus cone (same r and h = r) =

A. 4/3 πr^3

B. πr^3

C. $\frac{2}{3}\pi r^3$

D. $\frac{1}{3}\pi r^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

3581

QUESTION ID

S96-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{2}{3}\pi r^3$

GRADING NOTE $\pi r^3 - \frac{1}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

True or false?

A cone is ⅓ of the cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

3582

QUESTION ID

S96-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\pi r^3 - \frac{1}{3}\pi r^3 = \frac{1}{3}\pi r^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

3583

QUESTION ID

S96-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{2}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 4

Area of a ring • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer radius 5 and inner radius 3: exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

3585

QUESTION ID

S96-B4-Q1

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

True or false?

Ring area = $\pi(R^2 - r^2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5

3586

QUESTION ID

S96-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer 4 and inner 1:

A. 3π

B. 15π

C. 9π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

3587

QUESTION ID

S96-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15π

GRADING NOTE 16 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer r and inner y: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

3588

QUESTION ID

S96-B4-Q4

TYPE

Short answer

CORRECT ANSWER

$\pi(r^2 - y^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

True or false?

Ring area = $\pi(R - r)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

3589

QUESTION ID

S96-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\pi(R^2 - r^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 1

Use Cavalieri's principle for other solids • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A twisted tower and a straight prism both have height 50 m and every horizontal slice is a 10 m by 10 m square. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

3591

QUESTION ID

S96-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 5,000 m³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid with base area 36 and height 10 and a cone with base area 36 and height 10: compare volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

3592

QUESTION ID
S96-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Equal: 120 each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle shows that every pyramid with base area B and height h has volume ⅓Bh, once one such pyramid is known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

3593

QUESTION ID

S96-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal slices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

Two solids of height 8 have slice areas 5 and 5 at every level. Volumes?

A. 40 and 13.3

B. 5 each

C. Both 40

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

3594

QUESTION ID

S96-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Both 40

GRADING NOTE 5×8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two solids have equal heights, but at one level their slice areas differ. Can Cavalieri's principle be used? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

3595

QUESTION ID

S96-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the areas must be equal at every level.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 2

Model a dome as a hemispherical shell • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Find the volume of concrete to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

3597

QUESTION ID

S96-F2-Q1

TYPE

Short answer

CORRECT ANSWER

298.7 m³

GRADING NOTE Rounding to 0.1 required. $\frac{2}{3}\pi(10^3 - 9.5^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Concrete has density 2.4 t/m³. Estimate the dome's mass to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

3598

QUESTION ID
S96-F2-Q2

TYPE
Short answer

CORRECT ANSWER
717 t

GRADING NOTE Rounding to the nearest ton required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

True or false?

A shell's volume is the outer volume minus the inner volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3599

QUESTION ID

S96-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3599

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius 3 has volume

A. 9π

B. 27π

C. 36π

D. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3600

QUESTION ID

S96-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 18π

GRADING NOTE ⅔π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3600

ANSWER KEY Reduced copy of the question page, with its answer

A hemispherical shell has outer radius 10 m and inner radius 9.5 m. Why is its volume NOT $\frac{2}{3}\pi(10 - 9.5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5

3601

QUESTION ID
S96-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Volumes subtract, not radii: the shell is the difference of two hemispheres.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5